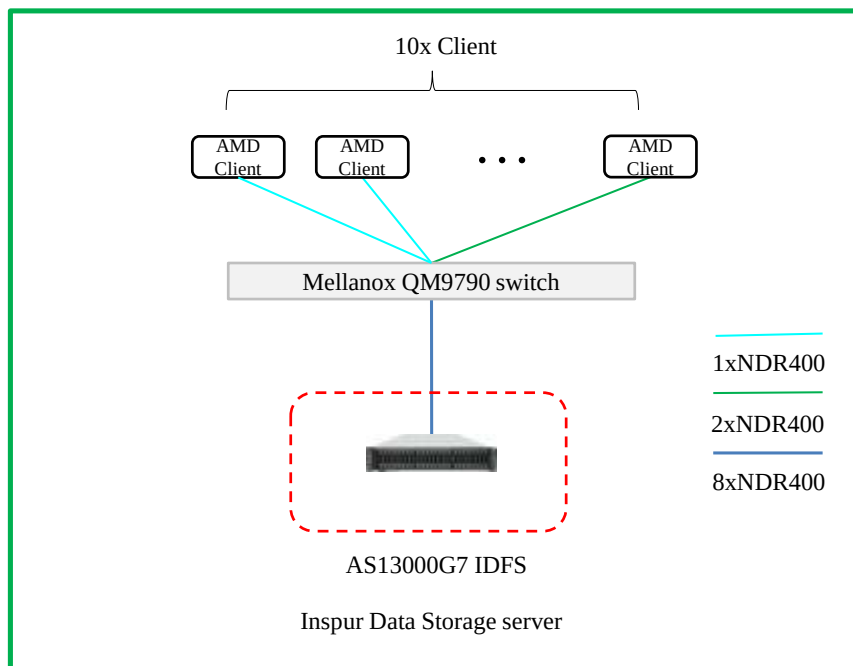

1. Overview

This benchmark test uses single chassis four-node system storage server, deploy the AS13000G7 IDFS distributed storage software.

2. Hardware Diagram



2.1 Client

This test utilized a total of 10 client nodes with three distinct hardware configurations as follows:

4 client nodes: Equipped with 1 AMD 9355 CPU, 768 GB memory, two 400 Gb IB network adapters, and one 1600 W power supply module each.

3 client nodes: Equipped with 1 AMD 9554 CPU, 384 GB memory, two 400 Gb IB network adapter, and one 1600 W power supply module each.

1 client nodes: Equipped with 1 AMD 9554 CPU, 512 GB memory, two 400 Gb IB network adapter, and one 1600 W power supply module each.

1 client nodes: Equipped with 1 AMD 9554 CPU, 512 GB memory, one 400 Gb IB network adapter, and one 1600 W power supply module each.

1 client nodes: Equipped with 1 AMD 9554 CPU, 384 GB memory, one 400 Gb IB network adapter, and one 1600 W power supply module each.

2.2 Server

The storage is one AS13000G7 device, with a height of 2U and four Nodes. It can be configured with 24×7.68TiB NVMe SSDs and 8×400Gb IB network cards, providing a unified distributed file system called IDFS.



2.3 Server Power Requirements

System component	Power supply unit	Nameplate rated power	Design Power
AS13000G7 Appliance 1	Power supply 1	2000W	8000W
	Power supply 2	2000W	
	Power supply 3	2000W	
	Power supply 4	2000W	
Totals		8000W	8000W

2.4 Switch

Mellanox QM9790 switch.

3. Software

3.1 Client

Client OS – Ubuntu 22.04

Mellanox OFED version – 5.8-4.1.5.0

3.2 Server

Storage Servers OS – kos release 5.8

Storage Servers Kernel – 4.19.91-26.6.25.kos5.x86_64

Mellanox OFED version – 24.04-0.6.6.0

Storage Servers version – idfs version 7.2